LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to other method of expressing positions i.e. pin points, range and bearing.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to other method of expressing positions i.e. pin points, range and bearing.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain and define a position on the earth.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Plotting range and bearing on chart; Safety: Handle chart paper carefully; Supervision: Instructor monitors plotting accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary methods used to define a position on the earth?
(b) Why is it necessary to use alternative methods besides latitude and longitude?
(c) Can you recall any military applications for range and bearing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Pin Point • Concept of Range • Concept of Bearing • Relationship between Range and Bearing • Importance in Navigation	L & Ds	PPT, IB	15 min
(b)	• Methods of expressing position • Pin point coordinates • Range measurement techniques • Bearing determination • Combined range and bearing usage	L & Ds	PPT, PDF	20 min
(c)	• Accuracy considerations • Error sources in measurement • Standard notation formats • Application scenarios • Comparison with Lat/Long	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Define a pin point in the context of expressing position.
(b) Explain how range and bearing are used together to locate an object.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of range and bearing will be utilized in the upcoming lesson on speed and velocity vectors.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to speed and velocity, vectors, wind velocity, vector addition and triangle of velocity.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min